

Comparison of the Prevalences and Antimicrobial Resistances of *Escherichia coli* Isolates from Different Retail Meats in the United States, 2002 to 2008

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Escherichia coli isolates were recovered from the National Antimicrobial Resistance Monitoring System retail meat program and examined for antimicrobial susceptibility. Retail meat samples (n = 11,921) from four U.S. states collected during 2002 to 2008, consisting of 2,988 chicken breast, 2,942 ground turkey, 2,991 ground beef, and 3,000 pork chop samples, were analyzed. A total of 8,286 *E. coli* isolates were recovered. The greatest numbers of samples contaminated with the organism were chicken (83.5%) and turkey (82.0%), followed by beef (68.9%) and pork (44.0%). Resistance was most common to tetracycline (50.3%), followed by streptomycin (34.6%), sulfamethoxazole-sulfisoxazole (31.6%), ampicillin (22.5%), gentamicin (18.6%), kanamycin (8.4%), amoxicillin-clavulanic acid (6.4%), and cefoxitin (5.2%). Less than 5% of the isolates had resistance to trimethoprim, ceftriaxone, ceftiofur, nalidixic acid, chloramphenicol, and ciprofloxacin. All isolates were susceptible to amikacin. Compared to beef and pork isolates, the poultry meat isolates had a greater percentage of resistance to all tested drugs, with the exception of chloramphenicol, to which pork isolates had the most resistance. More than half of the turkey isolates (56%) were resistant to multidrugs (≥ 3 classes) compared to 38.9% of chicken, 17.3% of pork, and 9.3% of beef isolates. The blaCMY gene was present in all ceftriaxone- and ceftiofur-resistant isolates. The cmlA, flo, and catI genes were present in 45%, 43%, and 40% of chloramphenicol-resistant isolates, respectively. Most nalidixic acid-resistant isolates (98.5%) had a gyrA mutation in S83 or D87 or both, whereas only 6.7% had a parC mutation in either S80 or E84. The results showed that *E. coli* was commonly present in the retail meats, and antimicrobial resistance profiles differed according to the animal origin of the isolates.